MIL-P-60045A(PA)
29 March 1977
SUPERSEDING
MIL-P-60045(MU)
30 May 1964

MILITARY SPECIFICATION

PROPELLANT M2 FOR USE IN 165MM CARTRIDGES

This specification is approved for use by Picatinny Arsenal, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers M2 Propellant for use in Cartridge, 165MM, HEP, M123A1.

1.2 Classification. The propellant shall be Type I, multi-perforated grains.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-652 - Propellants, Solid, for Cannons, Requirements and Packing

DRAWINGS

ARMY

8845054 - Charge, Propelling, 165MM Assembly

FSC: 1376

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Picatinny Arsenal, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

Title 49, Transportation, Parts 100-199

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402. Orders for the above publication should cite: 49 CFR 100-199 (latest revision)).

AMC PAMPHLET 706-114

Experimental Statistics, Section 5, Tables

(Application for copies should be addressed to the Publications and Reproduction Agency, Letterkenny Army Depot, Chambersburg, Pa. 17201).

3. REQUIREMENTS

3.1 Materials. The propellant shall be M2 Type I, conforming to the requirement of MIL-STD-652 and Dwg 8845054.

3.2 Proving ground assessments. The propellant when fired in the M135 Gun and with the weight of projectile (M123A1, 65.0 lbs.) for which it was manufactured, shall comply with the following requirements when tested as specified in 4.4.1.

3.2.1 Permissible individual maximum pressure. The propellant at recommended charge weight shall not produce a chamber pressure greater than 5,800 psi at any operating temperature (-50°F to +145°F) when fired in a standard weapon.

3.2.2 Charge establishment phase. At 70 degrees Fahrenheit (°F) and standard weapon conditions, the propellant shall be capable of assessment to a velocity of 850 feet per second (fps) (259.1 meters per second (mps)).

3.2.3 Uniformity phase

3.2.3.1 Uniformity of velocity. The standard deviation of velocity for the lot of 70°F shall not exceed 9.0 fps (2.7 mps).

3.2.3.2 Average pressure. At standard weapon conditions and recommended charge weight, the propellant shall not pro-

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duce an average pressure greater than 4,840 pounds per square inch (psi) at 70°F.

3.3 Workmanship. The best commercial practices shall be used in the formulation of propellant furnished under this specification, and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes. Unless otherwise specified, no reground propellant or nitrocellulose shall be used in the manufacture of propellant procured under this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. Quality Conformance Inspection shall be conducted on this item.

4.3 Quality conformance inspection

4.3.1 Inspection lot formation. Inspection lots shall comply with the lot formation provision of MIL-A-48078.

4.3.2 Examination and testing

4.3.2.1 Sampling (see 3.1 and 3.2). Each lot of propellant shall be sampled and tested for composition, form, dimensions and chemical/physical properties in accordance with MIL-STD-652. Each lot of propellant shall be sampled for proving ground tests in accordance with MIL-STD-652 except that ten containers shall be sampled for both initial and subsequent production lots.

4.3.2.2 Permissible individual maximum pressure. If at any time during testing, an individual pressure greater than that specified in 3.2.1 is observed, testing shall cease and the lot shall be rejected.

4.3.2.3 Charge establishment (see 3.2.2). If the lot does not meet the applicable requirement, testing shall cease and the lot shall be rejected.

4.3.2.4 Velocity uniformity (70°F) (see 3.2.3.1). If the product of the factor (see 6.3) times the velocity standard deviation of the rounds tested in accordance with 4.4.1.2 exceeds the applicable requirement, apply retest criteria in accordance with 4.4.1.3.

4.3.2.5 Pressure uniformity

4.3.2.5.1 Average pressure (see 3.2.3.2). If any average pressure obtained during initial or subsequent production at 70°F exceeds the applicable requirement, apply retest criteria in accordance with 4.4.1.3.

4.3.2.5.2 Individual pressure (subsequent production). If an individual pressure observed during the 70°F uniformity series, when corrected to standard conditions and re-

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commended charge weight, exceeds 5,800 psi minus the quantity 75 times the pressure temperature coefficient established during Initial Production Testing, and calibration round performance is acceptable (see 4.4.1.4), the lot shall be tested at +145°F in accordance with 4.4.1.2.1. If any individual pressure at 145°F, when corrected to standard conditions and recommended charge weight, exceeds 5,800 psi, testing shall cease and the lot shall be rejected.

4.4 Test methods and procedures

4.4.1 Proving ground assessments. These tests shall be conducted at a Government proving ground (designated by the contracting officer) in accordance with the applicable Proving Ground Acceptance Test Procedure. The following tolerances shall apply to all -50°F, 145°F and 70°F conditioning: -50°F from -50°F to -45°F; 145°F from 140°F to 145°F; 70°F from 67.5°F to 72.5°F.

4.4.1.1 Charge establishment

4.4.1.1.1 Charge slope method. Three or more groups of rounds loaded with selected charge weights of propellant (weights selected should bracket expected service charge weight) shall be conditioned and fired at 70°F. From the data obtained a charge weight shall be calculated which will yield the prescribed service velocity (see 3.2.2). This calculated charge weight shall then be loaded into rounds for the uniformity series (see 4.4.1.2). The difference between the corrected velocity obtained during the uniformity series and the service velocity shall be compensated for by a final adjustment in charge weight using the slope of the velocity charge weight curve previously established.

4.4.1.1.2 Constant slope method. After ten (10) lots of a given production series have been fired to provide a good knowledge of the propellant characteristics, the charge slope method of assessment may be discontinued and the charge weight determined by the constant slope method. A composite charge weight versus velocity curve can be established from test results generated. This curve can then be used as in 4.4.1.1.1 to calculate the charge weight needed to achieve service velocity. This calculated charge weight is then to be loaded into rounds for the uniformity series and final adjustment is to be made as indicated above for the charge slope method.

4.4.1.2 Uniformity

4.4.1.2.1 Initial production. Thirty rounds shall be assembled with the established charge weight of propellant. Ten rounds each shall be conditioned to -50°F, 70°F and 145°F (as indicated by thermocoupled control round). Rounds fired at 70°F shall be fired alternately with calibration rounds and shall be corrected to standard weapon/firing conditions (i.e. the difference between calibration performance under test conditions and standard conditions). Firing results

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at all temperatures shall be corrected to standard projectile weight (see 3.2) and to recommended charge weight conditions (i.e. the difference between as fired charge weight and recommended charge weight). The means and standard deviations of the velocities and pressures at all firing temperatures shall be calculated using corrected results. The average pressure-temperature coefficient for the lot shall be calculated from the pressure data obtained at 70°F and 145°F. The average of the coefficients for ten lots shall be utilized to calculate a 70°F upper pressure control for use in subsequent production.

4.4.1.2.2 Subsequent production. Ten (10) rounds shall be assembled and tested at 70°F, fired alternately with calibration rounds. The means and standard deviations of the velocity and pressure shall be calculated on results corrected to standard conditions and recommended charge weight. If any individual pressure exceeds the calculated 70°F upper pressure control (see 4.3.2.5.2), ten additional test rounds shall be conditioned and fired at 145°F. All results at 145°F shall be corrected to standard projectile weight (see 3.2) and recommended charge weight.

4.4.1.3 Retest criteria (see Figure 1). If for any reason the proving ground considers that test conditions have detrimentally affected the test results, additional charges as required shall be tested. A lot shall be retested whenever any of the following conditions exist and whenever the proving ground deems it necessary (see 6.8).

a. Test lot fails velocity standard deviation or average pressure at 70°F and calibration rounds fail one or more requirements stated in 4.4.1.4, retest for uniformity.

b. Test lot fails velocity standard deviation or average pressure at 70°F and any extreme test value is an outlier, retest for uniformity (see 6.9).

c. The final adjustment in charge weight exceeds two percent of the charge weight used in uniformity firings, retest for charge establishment in accordance with 4.4.1.1.1.

4.4.1.4 Referral criteria (see Figure 1). A test lot shall be referred to Picatinny Arsenal, ATTN: SARPA-QA-A-P for disposition whenever any of the following conditions exist and whenever the proving ground deems it necessary.

a. Calibration rounds fired in any test phase fail to comply with Section 3 of this specification.

b. Calibration correction applied exceeds two percent of the expected muzzle velocity.

c. Calibration components are not available.

d. Number of test or calibration rounds fired is less than specified.

e. Number of test or calibration results utilized in calculations is less than specified.

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5. PACKAGING

5.1 Packing and marking. Packing and marking shall be in accordance with MIL-STD-652, except that all propellant designated for use as reference shall be packed Level A.

6. NOTES

6.1 Ordering data. Ordering data shall be in accordance with MIL-A-48078(PA) with the following exceptions.

6.1.1 Procurement requirements. Procurement requirements of MIL-A-48078(PA) shall apply except for First Article Sample.

6.1.2 Contract data requirements. In addition to the normal distribution of records, when the propellant is procured by the Department of the Army, one copy of all ballistic test data and one copy of the Propellant Description sheet shall be forwarded to: Commander, Picatinny Arsenal, ATTN: SARPA-QA-A-P, and one copy to ATTN: SARPA-AD-E-P-2, Dover, New Jersey 07801.

6.2 Standard deviation. The standard deviation shall be calculated with (n-1) as the divisor in a standard statistical technique equivalent to that shown in MIL-STD-414, Section B, Example B-1.

6.3 Factors. The factors, as taken from Table A-21 of AMC Pamphlet 706-114 using the 95 percent upper confidence limits (A.05) and n-1 degrees of freedom, make allowances for the probability that a standard deviation from a smaller sample size exceeds the true standard deviation by chance alone.

6.4 Initial production lots. The initial production lots represent the first ten consecutive lots from each manufacturer which have been produced and met the acceptance criteria of this specification. Throughout production, if 3 consecutive lots fail on first test or lapse in production of 90 days or more occurs, testing shall revert back to initial production plan (see 4.4.1.2.1).

6.5 Surveillance sample. Samples shall be forwarded to Picatinny Arsenal, Dover, New Jersey, ATTN: SARPA-FR-G for the propellant surveillance test program in accordance with MIL-STD-652.

6.6 Test rounds. Test rounds shall be assembled in accordance with drawing 8845054.

6.7 Proving ground test summary

Initial Production

Temp (°F)	Sample Size	Assessed Velocity (fps)	Assessed Velocity (mps)	Lot Std Dev of Vel (fps)	Lot Std Dev of Vel (mps)	Perm Ind Max Pres (psi)	Avg Pres Max (psi)
-50	10	NA	NA	NA	NA	5,800	NA

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Initial Production (Cont)

Temp (°F)	Sample Size	Assessed Velocity (fps)	(mps)	Lot Std Dev of Vel (fps)	(mps)	Perm Ind Max Pres (psi)	Avg Pres Max (psi)
70	10	850	259.1	9.0	2.7	5,800	4,840
145	10	NA	NA	NA	NA	5,800	NA
		Subsequent Production					
70	10	850	259.1	9.0	2.7	see 4.3.2.5.2	4,840
145*	10	NA	NA	NA	NA	5,800	NA

*To be fired only when 70°F results indicate that marginal pressure performance may exist (see 4.3.2.5.2).

6.8 Retest. All original and retest data shall be submitted to Picatinny Arsenal, ATTN: SARPA-QA-A-P, Dover, New Jersey 07801.

6.9 Outlier. The following test shall be performed to determine if a single test result is an outlier. With a sample size of 10 at a significance level of 0.05, the critical value (T_c) is 0.477. Calculate a test value (T_t) as follows:

For Low Outlier

$$T_t = \frac{(X_2 - X_1)}{(X_9 - X_1)}$$

For High Outlier

$$T_t = \frac{(X_{10} - X_9)}{(X_{10} - X_2)}$$

Where:

X_1 = test result being tested as low outlier
X_2 = test result adjacent to X_1, after all the test results are ordered from smallest to largest
X_9 = test result adjacent to the largest test result
X_{10} = test result being tested as high outlier

X_1 or X_{10} is an outlier if T_t is greater than T_c.

6.10 Test anomalies. Any unusual occurrences during ballistic testing, such as excessive delays between rounds, stickers, poor projectile flights, etc., shall be reported in detail on the applicable firing record.

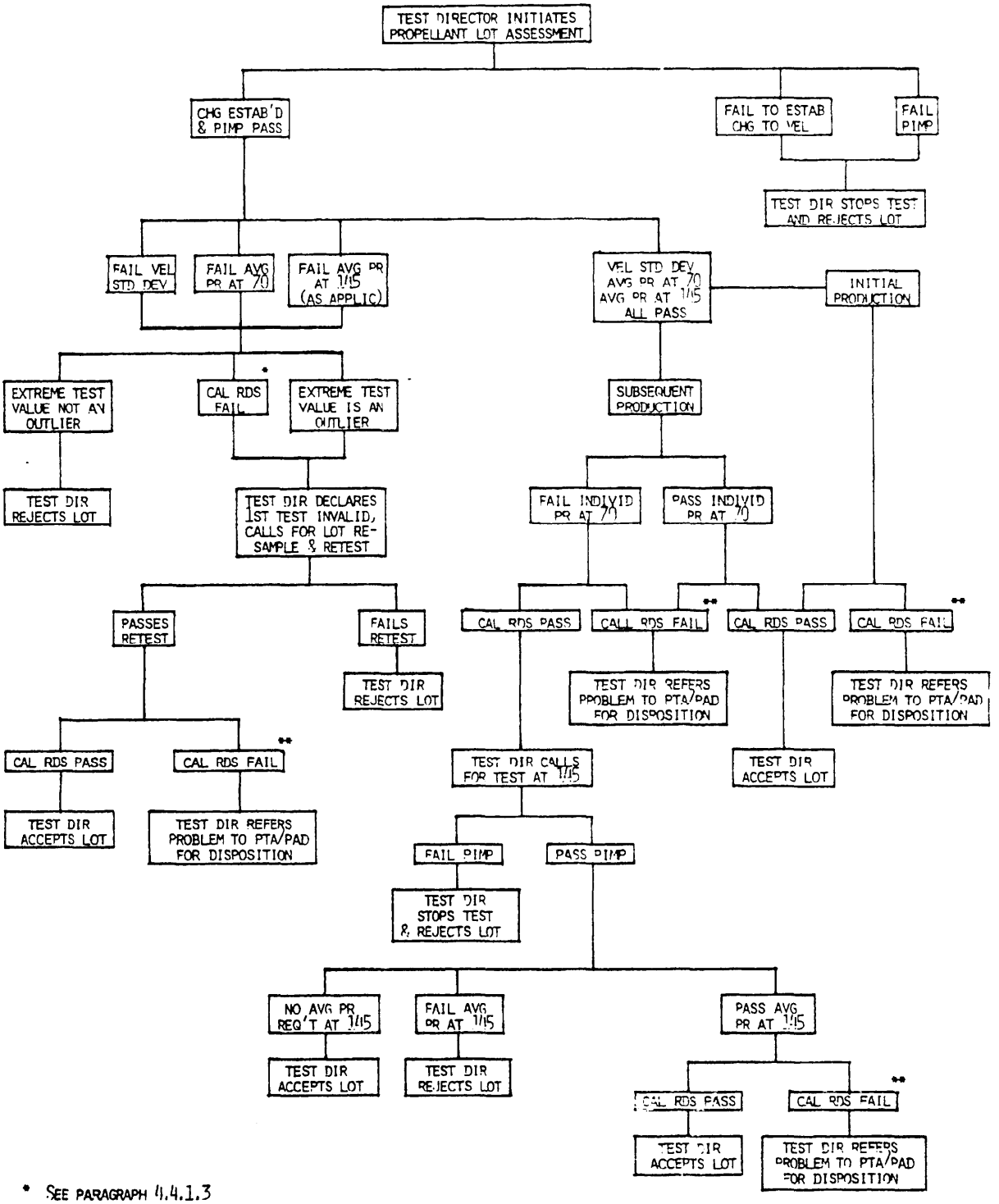
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FIGURE 1. DECISION PROCESS FOR PROPELLANT ASSESSMENT



* See paragraph 4.4.1.3

** See paragraph 4.4.1.4

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